Agenda

1. Exam Info Page - go.wisc.edu/1xxzfx
   * check for conflicts & sign up for makeup exams
   * register for McBurney Exams (scroll to bottom)
2. Getting Help:
   * Office Hours
   * MLC
   * Piazza
3. Trig Integrals (Section 7.2)
4. Trig Substitution (Section 7.4)

Integrals Involving Secant and Tangent

	The Integral	Use the Identity	Method
m odd	$\int \tan^m x \sec^n x \, dx = \int \tan^{m-1} x \sec^{n-1} x \sec x \tan x \, dx$	$\tan^2 x = \sec^2 x - 1$	$u = \sec x$ $du = \sec x \tan x \, dx$
n even	$\int \tan^m x \sec^n x \, dx = \int \tan^m x \sec^{n-2} x \sec^2 x \, dx$	$\sec^2 x = 1 + \tan^2 x$	$u = \tan x$ $du = \sec^2 x \, dx$
m even; n odd	$\int \tan^m x \sec^n x \, dx = \int (\sec^2 x - 1)^{m/2} \sec^n x \, dx$	$\tan^2 x = \sec^2 x - 1$	Use integration by parts

Example / Activities:

1.) $\int \tan^3 x \sec^4 x \, dx$

$u = \sec x$

$du = \sec x \tan x \, dx$

$\int \tan^2 x \sec^3 x \sec x \tan x$

$\int (\sec^2 x - 1)(\sec x)^3 \sec x \tan x$

$\int (u^2 - 1) u^3 \, du$

$\int u^3 - u^5 \, du$

$\left[\frac{u^6}{4} - \frac{u^4}{6}\right] + C$

$\boxed{\frac{\sec^4 x}{4} - \frac{\sec^6 x}{6} + C}$

2.) $\int \tan^2 x \sec^6 x \, dx$

$u = \tan x$

$du = \sec^2 x$

$(\tan^2 x - 1)$

$= \int \tan^2 x \sec^4 x \sec^2 x$

$= \int u^2 (1 + u^2)^2 \, du$

$= \int u^2 (u^4 + 2u^2 + 1)$

$\int u^6 + 2u^4 + u^2 \, du$

$\frac{u^7}{7} + \frac{2}{5} u^5 + \frac{u^3}{3} + C$

$\boxed{\frac{1}{7} \tan^7 x + \frac{2}{5} \tan^5 x + \frac{1}{3} \tan^3 x + C}$

3.) $\int \tan^2 x \sec x \, dx = \int (\sec^2 x - 1)\sec x \, dx = \int (\sec^3 x - \sec x) dx$

$= \int \sec^3 x \, dx - \underline{\int \sec x \, dx}$

Formula Sheet

$= \underline{\int \sec^3 x} - \underline{\ln|\sec x + \tan x|}$

example worked out in book

$= \underline{\left(\frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x|\right)} - \underline{\ln|\sec x + \tan x|} + C$

Using Product Identities

$$\sin(x)\cos(y) = \tfrac{1}{2}\left(\sin(x-y) + \sin(x+y)\right)$$

$$\sin(x)\sin(y) = \tfrac{1}{2}\left(\cos(x-y) - \cos(x+y)\right)$$

$$\cos(x)\cos(y) = \tfrac{1}{2}\left(\cos(x-y) + \cos(x+y)\right)$$

Example: $\int \sin(3x)\cos(2x) dx$

$= \frac{1}{2}\int \sin(3x-2x) + \sin(3x+2x)$

$= \frac{1}{2}\int \sin x + \sin 5x$

$= \frac{1}{2}\left[-\cos x - \frac{1}{5}\cos 5x\right] + C$

Trig Substitution (Section 7.3)

$a > 0$

Integrand contains	Substitution	Identity
$\int \sqrt{a^2 - x^2}\,dx$	$x = a\sin\theta,\ -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1 - \sin^2\theta = \cos^2\theta$ $a^2 - (a\sin\theta)^2 = (a\cos\theta)^2$
$\int \sqrt{x^2 + a^2}\,dx$	$x = a\tan\theta,\ -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$\tan^2\theta + 1 = \sec^2\theta$ $(a\tan\theta)^2 + a^2 = (a\sec\theta)^2$
$\int \sqrt{x^2 - a^2}\,dx$	$x = a\sec\theta,\ 0 \le \theta < \frac{\pi}{2}$ or $\pi \le \theta < \frac{3\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$ $(a\sec\theta)^2 - a^2 = (a\tan\theta)^2$

Example: $\int \frac{dx}{x^2\sqrt{4-x^2}}$

$$= \int \frac{2\cos\theta\, d\theta}{4\sin^2\theta \cdot 2\cos\theta}$$

$$= \frac{1}{4}\int \frac{d\theta}{\sin^2\theta}$$

$$= \frac{1}{4}\int \csc^2\theta\, d\theta = -\frac{1}{4}\cot\theta + C$$

adj / opp

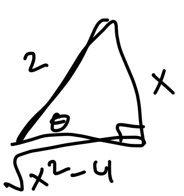
$$\boxed{-\frac{1}{4} \cdot \frac{\sqrt{4-x^2}}{x} + C}$$

$a^2 = 4$

$a = 2$

$x = 2\sin\theta \quad (\sin\theta = \frac{x}{2})$

$dx = 2\cos\theta\, d\theta$



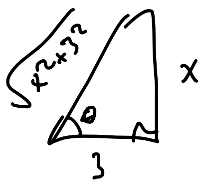
$c^2 = a^2 + b^2$

$2^2 = x^2 + b^2$

$b = \sqrt{x^2 - 4}$

$\frac{\sqrt{x^2-4}}{2} = \cos\theta \Rightarrow \sqrt{x^2-4} = 2\cos\theta$

Activity: $\displaystyle\int \frac{dx}{(x^2+9)^{3/2}}$ $\quad a^2 = 9$, $a = 3$



$x = 3\tan\theta$

$dx = 3\sec^2\theta$

$\tan\theta = \dfrac{x}{3}$

$c^2 = 3^2 + x^2$

$c = \sqrt{x^2+3^2}$

$\dfrac{\sqrt{x^2+9}}{3} = \sec\theta$ $\quad$ hyp / adj

$$\int \frac{\cancel{dx}\, 3\sec^2\theta}{(3\sec\theta)^3}\, d\theta = \int \frac{3\sec^2\theta}{27\sec^3\theta}\, d\theta$$

$$= \int \frac{d\theta}{9\sec\theta} = 9\int \cos\theta \, d\theta$$

$$= 9\sin\theta + C \quad \left(\sin\theta = \frac{op}{hyp}\right)$$

$$= \frac{9x}{\sqrt{x^2+9}} + C$$

Example: $\displaystyle\int (4x^2+9)^{1/2}\, dx$

Activity: $\displaystyle\int \frac{\sqrt{x^2-4}}{x}\, dx$

Example: $\int_1^3 \frac{dx}{\sqrt{x^2-1}}$